SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – SHORT ANSWER TEST (5 QUESTIONS)

Course Code:	ITA05[illegible]	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 1: Short Answer Test
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
Student Name:	I. Sarathy	**Reg. No.:**	192412599
Section / Batch:	B | 2024	**Date:**	5 | 10/26

Code of Conduct

I Sarathy I (192412599) *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Sarathy I

QUESTIONS: 5 **Total Marks:50**

Annexure A

Short Answer Test

Q1. Smart Traffic Monitoring System

A smart city is planning to install cameras at major traffic junctions to monitor vehicles, pedestrians, and traffic conditions. The captured images may vary in resolution, lighting, and camera position. **Apply the fundamental concepts of Computer Vision to design a suitable processing pipeline for this traffic monitoring system. Explain the major stages from image acquisition to object detection and classification.**

Q2. Automated Attendance System

A college wants to implement an automated attendance system using cameras that capture students sitting inside classrooms. The images may contain different lighting conditions, face orientations, shadows, and multiple students.

Analyze how image representation, preprocessing, feature extraction, and face recognition can be applied to develop the proposed attendance system. Discuss the major challenges that may affect its accuracy.

Q3. Medical Image Analysis System

A hospital plans to use a Computer Vision system to assist doctors in analyzing X-ray images for identifying abnormal regions. The images may contain noise, poor contrast, and different brightness levels.

Evaluate the importance of image resolution, intensity, preprocessing, segmentation, and feature extraction in this application. Explain how these factors can influence the accuracy and reliability of the system.

Q4. Security Surveillance System

A shopping mall wants to develop an intelligent surveillance system using cameras installed at entrances, corridors, and parking areas. The system must identify people and objects under different lighting and crowded conditions.

Analyze how the fundamental stages of a Computer Vision system can be used for this application.

Explain the effects of illumination, occlusion, camera position, and image quality on the final results.

Q5. Smart Agriculture Monitoring

A farming organization wants to use drone cameras to identify healthy plants, damaged plants, weeds, and disease-affected areas in large agricultural fields. The captured images may contain shadows, different lighting conditions, and variations in plant size.

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, in-depth4, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: Date: 5/10/26	Signature: ______________ Date: ______________	Signature: ______________ Date: ______________

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ITA0510 - Computer Vision

CO1 - AT1 - Short Answer Text

mart Traffic Monitoring System:

Image Acquisition → preprocessing → Segmentation → Feature
traction → Detection → Classification.

★ Image Acquisition:

Capture traffic images using cameras.

★ preprocessing:

Resize, remove noise and improve brightness.

★ Segmentation:

Separate vehicles and pedestrians from background.

★ feature extraction:

Extract shape, size, color and edges.

★ Detection & classification:

Detect and classify cars, buses, bikes and pedestrains

★ output:

Traffic count and traffic condition

) Automated Attendance System:

Image Acquisition → preprocessing → Face Detection
→ Feature Extraction → Face Recognition

★ Image Representation

Images are represented as pixels / RGB or grayscale

★ preprocessing

Remove noise and adjust brightness

★ Feature Extraction

Extract facial features

★ Face Recognition

Compare faces with registered student data.

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Attendance:

Mark recognized student as present.

3) Medical Image Analysis System.

Resolution:

Higher resolution gives more medical details.

Intensity:

Shows brightness differences between tissues

preprocessing:

Removes noise and improves contrast

Segmentation:

separates abnormal regions

Feature Extraction:

Extract shape, texture and intensity

classification:

Identifies normal and abnormal regions.

4) Security Surveillance System

- ☆ Cameras capture people and objects.
- ☆ preprocessing improves image quality.
- ☆ Detection identifies people and objects.
- ☆ features such as shape and size are extracted.
- ☆ classification identifies objects and suspicious activities.

5) Smart Agriculture monitoring.

- ☆ Drone captures captures field images.
- ☆ Removes noise and corrects lighting.
- ☆ Separates plants, Soil and weeds.
- ☆ Extracts color, Shape and texture.
- ☆ Identifies healthy, damaged and diseased plants.
- ☆ Helps farmers locate diseases and weeds.